INCH-POUND

A-A-54772
31 December 1992

COMMERCIAL ITEM DESCRIPTION

STETHOSCOPE, Combination Type, Light Weight

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-S-36588A which is cancelled.

This commercial item description covers requirements for lightweight, bell-diaphragm, combination stethoscopes suitable for diagnostic analysis.

Salient characteristics:

Scope. Stethoscopes covered by this document shall be of the following sizes:

Size 1 - Adult
Size 2 - Child

Requirements. The stethoscope shall essentially consist of a binaural section with tubing and a chestpiece. Each stethoscope shall be fully assembled and supplied with two pairs of ear tips and two extra diaphragms. The stethoscope shall be approximately 24 inches in length. The weight of the stethoscope shall not exceed 4.6 ounces.

Binaural section. The binaural section shall consist of two aural tubes, a steel spring, tubing and two ear tips.

The aural tubes shall be fabricated of chromium plated or chromium over nickel plated brass tubing. The proximal end of each tube shall have external threads to receive and securely hold the ear tips. The distal ends of the tubes shall have suitable means for securely retaining the plastic tubing. The aural tubes shall be held together by means of a steel spring. With the steel spring assembled, the aural tubes shall be rotationally adjustable. The steel spring shall be "U" shaped and fabricated of corrosion resistant steel or carbon steel with chromium or chromium over nickel plating.

AMSC N/A FSC 6515

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The tubing shall be a one piece construction of "Y" shape made of polyvinyl chloride plastic. The vertical portion of the tubing shall be at least 13 inches long. All three ends of the tubing shall make leakproof connection with their attached parts. The tubing shall be smooth, flexible, seamless and not tacky to the touch. The tubing shall not kink or exhibit any sign of permanent deformation with repeated use.

The ear tips shall be made of suitable plastic or rubber. The tips shall be a minimum of 1/2 inch long, and be provided with a lumen for the entire length. The lumen shall have well defined internal threads that will fit securely over the threads of the aural tubes The diameter of the ear tip shall be 17/32 ± 1/32 inch.

Chest piece. The chest piece shall be a combination type, embodying a bell and diaphragm unit, and shall operate with a two position valve that will allow either bell or diaphragm as a functional unit. Appropriate marking or other indication shall be present on the chest piece to indicate the function selected.

The combination chest piece shall substantially reduce all sounds for the chest piece not selected so that a minimum increase of 25 decibels (db) is required for audible sound to pass through that chest piece, when tested as specified.

The chest piece assembly shall consist of a body, retaining ring and diaphragm.

The chest piece body shall be fabricated of suitable metal with mirror finish chromium over nickel plating. The diaphragm side shall be suitably threaded to receive the retaining ring. The diaphragm shall be held in place by the inner circumference of the ring. The diameters of the diaphragm and bell shall be as follows:

	Bell	Diaphragm
Size 1	1-3/32 inches minimum	1-1/2 to 2 inches
Size 2	1 inch minimum	1 to 1-1/2 inches

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Sound transmission. The sound transmission shall be such that the corrected threshold values in decibels (lowest audiometer volume for audibility) shall be the following maxima when tested as specified:

	CYCLES PER SECOND				
Frequency	125	500	1,000	2,000	3,000
Bell	0	5	10	-*	-
Diaphragm	-	-	10	15	35

* Blank spaces indicate absence of requirement.

Tests.

(a) Effectiveness of selector valve. This test shall be run at 125 cycles per second (cps) only. The bell side of the chest-piece is shut off. The stethoscope is fitted to the ears of the tester and the binaurals adjusted for optimum hearing. The bell is placed concentrically over the output earphone and an airtight junction established. The diaphragm (on top side) is masked with a sponge or the palm of the hand. The volume of sound is increased until the threshold value is obtained. There shall be a minimum increase of 25 db required to reach the threshold value with the bell shut compared to the threshold value at 125 cps with the bell open. Since this determination is based on the difference in readings, no corrections should be applied. The greater the increase in dbs required to reach the audibility threshold, the greater the effectiveness of the selector valve.

(b) Sound transmission test All sound transmission tests on the stethoscopes shall be conducted in a room with a minimum and constant level of background noise. An audiometer complying with ANSI Standard S3.6 shall be the test instrument.

The audiometer is allowed to warm up for 15 minutes and hearing of the tester shall be established as being normal by comparing his audiogram with audiograms of six other people whose hearing is considered normal. Normal hearing is defined as being no more than 15 db above audiometric zero for a given frequency. The frequencies at which the tester's hearing is screened are at 125, 500, 1,000, 2,000, and 3,000 cps. Audibility thresholds are determined for each frequency, that is, the minimum db level at which sound is detectable. These threshold values for the tester are

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run each time just prior to the evaluation of stethoscopes. Audiogram values for the tester shall be run each half-hour in order to compensate primarily for variations in the background noise level. The maximum audiogram values at each frequency, whether obtained with one ear or the other during the testing, shall be used as corrections. Although both earphones are placed over the ears of the tester, only one earphone shall transmit the sound to the tester. The other earphone is kept soundless. The audiogram values are subtracted from the corresponding frequency values obtained on the stethoscopes to give corrected readings.

Frequency response characteristics. The operative side of the chest piece is placed in the center of the output earphone of the audiometer A pressure on the chest piece just adequate to maintain a hermetic seal is maintained during the testing. The binaurals of the stethoscopes are adjusted for optimum reception by the tester. When the chest piece, especially the bell of the pediatric size, is smaller than the center hole of the foam rubber pad of the audiometer earphone, it will be necessary to use a soft, one-eighth inch layer of foam rubber or plastic with a hole in it just slightly smaller than the bell being tested. Such an adapter gasket provides the necessary hermetic coupling of the bell to the audiometer earphone. Starting with a 125 cps frequency setting the volume of sound is varied until the point is reached at which the sound from the audiometer is just detectable. This is the threshold value. The same procedure is repeated for 500, 1,000, 2,000 and 3,000 cps. The audiogram value for each frequency is subtracted from the value for the corresponding frequency obtained on the stethoscope being tested. The resultant corrected readings are compared with the requirements stated under salient characteristics.

Identification marking. The chest piece shall be permanently and legibly marked with the manufacturer's name or registered trademark.

Workmanship. The stethoscope shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One stethoscope of the size specified, constitutes one unit.

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QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for determination of two or more test characteristics.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified herein. Unit of product for test purposes shall be one stethoscope. Inspection level shall be S-1, and acceptance number shall be zero.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

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If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

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NOTE: The following National Stock Numbers, Item Identifications and Sizes are covered by this document:

NATIONAL STOCK NUMBERS	SIZES	ITEM IDENTIFICATIONS
6515-00-935-4088	1	STETHOSCOPE, Combination Type, Lightweight, Adult
6515-00-935-4089	2	STETHOSCOPE, Combination Type, Lightweight, Child

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project Number: 6515-5296
Location: c:\word\gandhi\stethcid

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given.

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER A-A-54772	2 DOCUMENT DATE (YYMMDD) 31 December 1992

3 DOCUMENT TITLE

STETHOSCOPE, Combination Type, Light Weight

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)

5 REASON FOR RECOMMENDATION

6 SUBMITTER

a [illegible]	b ORGANIZATION
c ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable) / 7 DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME Defense Personnel Support Center ATTN: DPSC-MSE	b TELEPHONE (Include Area Code) (1) Commercial (215)7372870 (2) AUTOVON 444-2870
c ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike Suite 1403 Falls Church VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete*

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